Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/09/2025
PI Name:	Friedman, Michael Ph.D.		
Project Title:	Genetic Variability in Microgravity-Induced Bone and Muscle Loss		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRI-RFA-1802-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	01/01/2019	**End Date:**	12/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per TRISH (Ed., 11/2/2020) 'NOTE: End date changed to 12/31/2020 per March 2020 report from TRISH (Ed., 4/7/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Henry J Ph.D. (MENTOR: Virginia Commonwealth University)		
Grant/Contract No.:	NNX16AO69A-P0301		
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Task Description:	POSTDOCTORAL FELLOWSHIP Unloading of limbs due to prolonged exposure to microgravity, as would occur with space travel, decreases bone and muscle volume, and strength, making bone and muscle more susceptible to injury. Countermeasures to prevent these decreases need to be developed for long-duration spaceflight, for instance to Mars. The underlying mechanisms that lead to bone and muscle loss from unloading must be better understood to develop effective countermeasures. Interestingly, as much as 80% of the variability in bone mass in humans is due to genetic factors. In mice, the response to changes in mechanical loading is mouse-strain dependent. This suggests that the response of bone and muscle to unloading may be dependent on genetics. However, it remains unclear how genetic diversity affects the response of bone and muscle to unloading. One approach to study the effects of genetic variation on the response of bone and muscle to unloading is through genetic analysis in Diversity Outbred (DO) mice. The DO is a population of outbred mice derived from eight genetically different founder strains. DO founder strains can be studied as a first step to evaluate the effects of genetic variation on the response of bone and muscle to unloading. These mice are a uniquely powerful animal model for identifying genes that drive the response to interventions. Bone and muscle interact to simultaneously adapt to changes in mechanical loading. Both tissues respond via similar gene pathways to decrease tissue volume and strength in response to unloading. Growing evidences suggests bone and muscle cells interact through biochemical signaling independent of loading. Further research is needed to elucidate the effects of genetic variation on these signaling pathways, especially in response to unloading. This project proposes to utilize mice of different genetic backgrounds and an animal model of unloading via simulated microgravity to examine effects of genetic variation on bone and muscle loss in response to unloading. Global Hypotheses: Genetic differences between mouse strains result in a differential response to mechanical unloading in bone and muscle. Moreover, mouse strains that have bones and muscles less sensitive to unloading harbor genetic differences resulting in variable activation levels of known gene pathways involved in bone and muscle mechanical adaptation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research shows that genetic variability can cause some individuals to be protected from the negative effects of unloading experienced during spaceflight or immobilization. This is the first study to look at how both bone and muscle respond to unloading in mice of different genetic backgrounds, comparing different models of unloading.
Task Progress:	POSTDOCTORAL FELLOWSHIP 1. Original project aims/objectives Unloading of limbs due to prolonged exposure to microgravity, as would occur with space travel, decreases bone and muscle volume, and strength, making bone and muscle more susceptible to injury. Countermeasures to prevent these decreases need to be developed for long-duration spaceflight, for instance to Mars. The underlying mechanisms that lead to bone and muscle loss from unloading must be better understood to develop effective countermeasures. Interestingly, as much as 80% of the variability in bone mass in humans is due to genetic factors. In mice, the response to changes in mechanical loading is mouse-strain dependent. This suggests that the response of bone and muscle to unloading may be dependent on genetics. However, it remains unclear how genetic diversity affects the response of bone and muscle to unloading. One approach to study the effects of genetic variation on the response of bone and muscle to unloading is through genetic analysis in Diversity Outbred (DO) mice. The DO is a population of outbred mice derived from eight genetically different founder strains. DO founder strains can be studied as a first step to evaluate the effects of genetic variation on the response of bone and muscle to unloading. These mice are a uniquely powerful animal model for identifying genes that drive the response to interventions. Bone and muscle interact to simultaneously adapt to changes in mechanical loading. Both tissues respond via similar gene pathways to decrease tissue volume and strength in response to unloading. Growing evidences suggests bone and muscle cells interact through biochemical signaling independent of loading. Further research is needed to elucidate the effects of genetic variation on these signaling pathways, especially in response to unloading. This project proposes to utilize mice of different genetic backgrounds and an animal model of unloading via simulated microgravity to examine effects of genetic variation on bone and muscle loss in response to unloading. Global Hypotheses: Genetic differences between mouse strains result in a differential response to mechanical unloading in bone and muscle. Moreover, mouse strains that have bones and muscles less sensitive to unloading harbor genetic differences resulting in variable activation levels of known gene pathways involved in bone and muscle mechanical adaptation. 2. Key findings Unloading hindlimbs of the 8 DO mice founder strains via single limb immobilization or hindlimb suspension results in significantly different magnitudes of loss of bone mass, bone strength, and muscle mass. In bones, unloading caused upregulation of genes associated with bone formation after 3 weeks. Changes in gene expression were mouse strain-dependent. In muscles, unloading caused mouse strain-dependent changes in markers of muscle atrophy and protein synthesis. Both hindlimb suspension and single limb immobilization result in similar musculoskeletal effects. Single limb immobilization can be a great model of unloading where one limb is unloaded and the contralateral limb is used as an intra-animal control. This reduces the number of animals needed for detecting significant effects of unloading and increases reproducibility of results by reducing variance by having intra-animal controls. Some of the mouse strains we used did not adapt to the hindlimb suspension restraints very well and had a high rate of attrition, requiring more mice to be used to get significant effects. Single limb immobilization can model bone and muscle loss from unloading, using fewer mice and with less risk of death. 3. Impact Genetic differences cause some bones and muscles to be protected from the negative effects of unloading. Single limb immobilization will be a great model for studying effects of unloading in genetically unique DO mice. Further work

	needs to be done to identify specific genes responsible for the different responses to unloading. We will continue this research in the Go For Launch Award. [Note: See "Musculoskeletal Response to Disuse in a Genetically Diverse Mouse Population", PI: Friedman (Ed., 7/9/25).] 4. Proposed research plan for the coming year We will continue analysis of samples collected from unloading via hindlimb suspension on the 8 DO mice founder strains. Bone and muscle samples will be analyzed and compared to bone and muscle samples we have in our lab from female Balb/C spaceflight mice and hindlimb suspension mice. The results will be compared to results of single limb immobilization. Results will be submitted to journals for publication. The first of approximately 4 articles related to the TRISH fellowship will be submitted in 2022.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Friedman MA, Buettmann EG, Zeineddine Y, Abraham LB, Hoppock GA, Meas SJ, Zhang Y, Farber CR, Donahue HJ. "Genetic variation influences the skeletal response to hindlimb unloading in the eight founder strains of the diversity outbred mouse population." J Orthop Res. 2024 Jan;42(1):134-140. https://doi.org/10.1002/jor.25646 ; PubMed PMID: 37321985; PubMed Central PMCID: PMC10721729 , Jan-2024
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